

Embedded Systems Project 2022-23

Final Report

Title: ESP Buggy Final Report

Group Number: 25

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1. Executive Summary, Introduction

1.1. Executive Summary

This report comprehensively analyses a yearlong project focused on designing and constructing an autonomous line-following buggy. The primary objective of the project was to develop a battery-powered buggy capable of autonomously navigating a track, climbing a ramp with a maximum incline of 18° , executing a 180° turn using a (Bluetooth Low Energy) BLE command, and coming to a controlled stop at the end of the lined track when the line terminates. The report critically examines the challenges encountered during the project. It discusses the strategies employed to address these challenges while providing insights into the group's approach to fulfilling the project's objectives.

The team achieved the project's goals by implementing various design and technical features. The compact design of the buggy, along with the incorporation of three Proportional-Integral-Derivative (PID) Controllers, an appropriate gearbox, and a line sensor Printed Circuit Board (PCB) with hardware features, significantly contributed to the buggy's autonomous capabilities. These features facilitated line break avoidance and ensured reliable signal processing from the TCRT5000 sensors.

However, a recurring mechanical issue with the left wheel posed a significant challenge throughout the project. The wheel exhibited problems such as loosening, failure to rotate, and uneven rotation, adversely affecting the buggy's performance during the final technical demonstration. Adjustments had to be made to the code on the day of the demonstration to accommodate the new wheel, as it did not respond as expected with the previous code. Unfortunately, the team could not test the revised software on an official track before the race due to time constraints, resulting in the buggy's inability to complete the track in the timed runs. Nonetheless, the team made further adjustments, and the buggy completed the track in a subsequent non-timed run.

Throughout the project, the team maintained a structured and organised approach. Clear objectives and plans were established early on, and tasks were assigned based on individuals' strengths, interests, and availability to optimise productivity. Effective communication played a pivotal role in the team's success, with various social media platforms utilised for coordination, file-sharing platforms employed for seamless collaboration, and regular in-person meetings conducted.

This report documents the project's progress and outcomes and critically analyses the encountered challenges and the corresponding resolutions. It highlights the innovative features incorporated into the buggy's design and the technical expertise demonstrated in implementing a robust software system. The report further reflects on the team's ability to adapt and problem-solve in a dynamic project environment, showcasing resilience and dedication.

In conclusion, despite the mechanical issue with the left wheel impacting the buggy's performance during the timed runs, the project showcases the team's strong technical competence and commitment to achieving the set objectives.

1.2. Introduction

This report comprehensively analyses a yearlong project focused on designing and constructing an autonomous line-following buggy.

The project's primary objective is to develop a battery-powered buggy capable of:

- Autonomously navigating a lined track.
- Ascending and descending an 18° incline at a constant speed.
- A 180° turn execution.
- Controlled stopping at the end of the line.

The report aims to thoroughly examine the project's challenges, strategies, and outcomes, offering valuable insights for future developments in this rapidly evolving field. A structured and organised approach, emphasising effective communication, task delegation, and utilising individual strengths, characterised the methodology employed in this project. Collaborative efforts were facilitated through various communication platforms, file-sharing sites, and regular in-person meetings, ensuring productive teamwork throughout the project.

Structured to provide a detailed account, the report will delve into the project's objectives, approach, technical challenges encountered, innovative solutions devised, and overall achievements. Figure 1 shows a graphic of the final buggy. It illustrates some hardware features, including the sensor shield and array. Figure 2 illustrates the CAD model for the buggy without the sensor shield or second castor wheel.

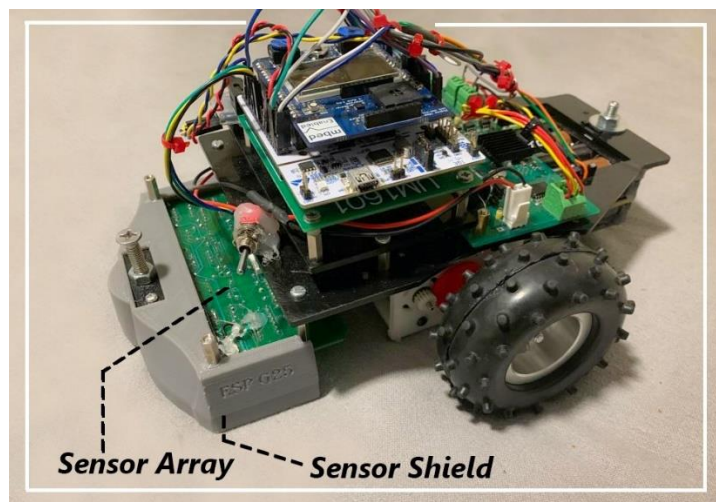


Figure 1 - Final design of the buggy



Figure 2 - CAD image of buggy design

2. Final System Components Summary

The buggy comprises several components working harmoniously to create an operational system. This section summarises the components used on the buggy and how they contributed to achieving the project objectives, as stated in [1].

2.1. Mechanical Components

The buggy incorporates various mechanical components such as fasteners, washers, screws, nuts, gearboxes, castor wheels and rubber tyres. These components were strategically employed to enhance the robustness of the buggy chassis, rendering it resilient against physical impacts and high-intensity vibrations resulting from the motor operation.

To traverse straight tracks and inclined slopes, the motors necessitated adequate torque generation to propel the buggy's weight and the additional resistive forces caused by gravity. Utilising experimental data obtained during the motor characterisation lab [2], it was determined that a torque of 0.12 Nm was required.

To attain the torque for inclines, Gearbox 2, featuring a 15:1 Gear ratio, was selected to amplify the torque. This selection offered the advantage of increased speed on straight tracks, albeit resulting in relatively slower performance on inclines. However, this drawback was overcome by implementing a speed controller, which demanded a minimum current of 1.5 A. It is worth noting that while there is an incline on the track, it constitutes only a minor portion of the track. The benefits in terms of speed outweighed the associated cons, despite the risk of the motor stalling and overheating.

The castor wheels were crucial as near-frictionless supports for the buggy's weight. The motor shaft was used as a pivot to ensure stability during movement and minimise drag torque on the wheels. Furthermore, the distribution of the components was optimised to minimise the reaction forces on the tyres and maximise the forces on the castor wheels, effectively reducing the overall friction in the system. This choice facilitated increased speed and reduced power requirements for the buggy. The final mechanical components consisted of rubber tyres, specifically chosen to enhance traction and prevent wheel slippage while the buggy was in motion.

2.2. Electronic Components

The buggy's electronic components comprised a Motor drive board, an HM10 Module, Hall effect encoders, Brushed DC Motors, an STM32 Microcontroller, a Sensor Array PCB, and Batteries. The batteries served as the power source for all electrical components and were controlled using an SPDT switch. The HM10 Module facilitated the acceptance of BLE commands from compatible devices, allowing for state switching on the buggy, particularly during turn-around interventions. The Brushed DC Motors provided the necessary mobility for the buggy and played a crucial role in the line following by contributing angular velocity through rotation.

The Hall Effect Encoders employed the Hall Effect to measure the wheel rotations, enabling the quantification of each wheel's velocity. This data served as feedback for the PID speed controller, contributing to precise speed control. Acting as the brain of the buggy, the STM32 Microcontroller played a central role by facilitating the transmission and reception of electrical signals and coordinating the actions of the various electrical components.

The Motor Drive Board Served as the Intermediary between the Motors and the Microcontroller. The Microcontroller alone could not supply sufficient power to drive the motors, so the Drive Board was necessary. It allowed the direction to be controlled through an H-bridge, speed control through PWM signals and mode control through a digital signal. A drawback of the drive board was its inversion of PWM signals, resulting in a short-lived overshoot upon initial activation. The controller design addressed this challenge, detailed in Section 2.4 of the report.

The Sensor Array PCB played a crucial role by providing an electrical signal representing the buggy's position relative to the white line. The PID steering controller used this feedback to implement a closed-loop controller. The PCB comprised various passive and active components such as resistors, a ULN2003A Darlington Buffer, 6 TCRT5000 infrared (IR) sensors, and op-amps. The TCRT5000 was selected for its high voltage output contrast between the white and black track and its built-in visible light filter, which enhanced its effectiveness in accurately detecting the white line [3].

2.3. Control System

Three individual PID controllers were operating concurrently on the buggy. One for each motor (i.e., the wheels) and one to control the steering of the buggy. The Motor PIDs were tuned by defining the motor system as a second-order transfer function using speed data obtained from each motor in MATLAB. The model was connected to a closed-loop PID controller in Simulink, where the Controller library was used to get the PID parameters to achieve critical dampening.

A similar process was adopted for the steering control, with the distinction that the model was developed based on the system's dynamics rather than measured data.

The modelling process was quick, and the simulated response matched the real-world response with an accuracy of over 70%, up to the maximum speed of 1.4 m/s (max speed of the buggy on a flat track). It is worth noting that the speed limit corresponds to the maximum velocity at which the motors can propel the buggy without an electrical failure.

2.4. Software Components

Software is one of the most essential components of the buggy. Without it, line following would still be implemented using purely electrical circuits, but it would increase the cost and the amount of time required to achieve similar performance.

The software was written using C++ on a remote virtual platform called Mbed. Objects Oriented programming was used in the programming architecture to improve readability, reduce redundancy, and make the code easier to debug. A drawback of this is that it takes longer for programs to execute. However, the STM32 can handle large computations and a 3-stage pipeline, making this problem minor to this project.

The objectives of the software were to implement the PID controllers by receiving and sending signals to the motor drive board and sensor array. This was implemented by discretising the PID equation and transforming it to code. It also needed to turn the buggy around when it received a Bluetooth command and continue following the line after. Finally, it was required to detect the end of a line and automatically stop within 20 cm from detection.

A state machine was created to achieve these purposes, and the state register was determined using the BLE command. A Bluetooth interrupt was used instead of polling to ensure turn-around action occurred semi-instantaneously.

2.5. Changes from the original design

Some changes were made to the design of the buggy.

- Due to a design failure, the weight of the buggy was not distributed solely on the castor wheels in the back, and as a result, a second castor wheel was placed at the front of the buggy to prevent it from tilting when moving.
- The distance between the sensor and the ground was increased, as the previous distance was too close to the ground, producing errors in the PCB output.
- A sensor shield was added to the chassis to prevent direct light and remove the effects of ambient light on the sensor readings.

2.6. Performance Analysis

The performance evaluation of the autonomous line-following buggy revealed several noteworthy achievements. The following points highlight the successful components and functionalities of the buggy:

- **Speed Controller:** The speed controller demonstrated excellent performance by effectively dampening the speed response of the buggy. It successfully addressed the initial overshoot issue caused by the PWM signal inversion of the motor drive board. Consequently, the buggy maintained a consistent speed while navigating inclines and descents, ensuring smooth and controlled movement. Figure 3 shows the measured critical damping of the buggy speed at 1.4 ms^{-1} .

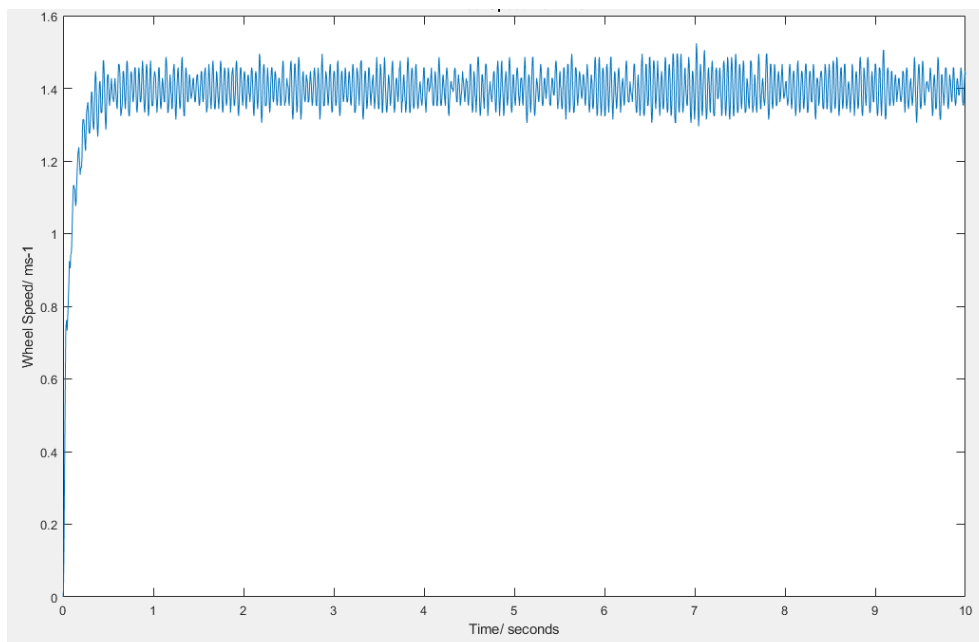


Figure 3 - Speed Controller Response

- **Steering Control:** The steering mechanism exhibited remarkable precision and responsiveness, effectively following curves with minimal deviation. The system demonstrated a slight overshoot but rapidly converged within a second.

Even at higher speeds, the steering controller maintained its stability and accuracy, enabling the buggy to manoeuvre through the track quickly. Figure 4 illustrates the closed-loop step response of the buggy's line-following model at a target speed of 1.4 ms^{-1} .

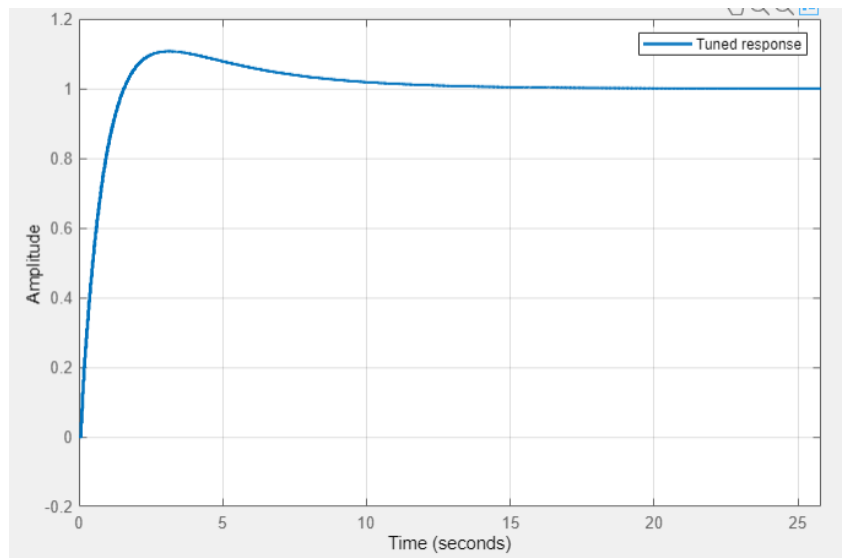


Figure 4 - Step Response of steering controller

- **Turn-Around Command:** By maintaining the activation of the speed controller during the turn-around manoeuvre, the buggy achieved synchronised wheel speeds. This synchronisation facilitated the smooth execution of a 180° turn around its vertical axis, showcasing the efficacy of the implemented control system.
- **Line Breaks and Controlled Stop:** The buggy's functionality included the ability to ignore line breaks and halt at the end-of-line accurately. This feature ensured that the buggy ignored interruptions in the line while consistently achieving the project's objective of coming to a controlled stop at the designated endpoint.

The successful components and functionalities highlight the significant achievements of the autonomous line-following buggy. These accomplishments underscore the efficacy of the implemented speed controller, steering control, turn-around command, and line detection mechanisms. By excelling in these areas, the buggy demonstrated its capability to complete the project's objectives and operate reliably throughout the designated track.

2.7. Extra Features

Several features were added to improve the performance of the buggy.

A running average was set up for the wheel speed measurement. It acted as a low pass filter (LPF) and removed noise due to random error.

A filtered derivative was used in the PID controller for the steering. This acted as a low-pass filter and largely ignored high-frequency noise in the data from the sensor board. This improved the stability of the line-following.

An energy monitor was set up to indicate when the batteries required charging by turning the LED on the application shield blue. This helped to ensure that the buggy had enough power for all demonstrations.

Op-amp circuitry and extra power ports were added to the sensor array PCB. The circuitry took the values from the 6 IR sensors and used them to calculate the error term for the steering control. It reduced the required analogue pins and reduced the delay due to the sampling time of the Analogue to Digital Converter (ADC).

Finally, the sensors were arranged in a zig-zag pattern to overcome the line breaks on the track.

3. Team Organisation and Planning

3.1. Summary of the Project Objectives

The project objectives as initially outlined in the proposal report [1]. Remained consistent throughout the duration of the project. The objectives encompassed the following milestones to be achieved in each semester.

Semester 1:

- Designing and developing an autonomous line-following buggy capable of accurately and efficiently following a predefined path.
- Equipping the buggy with sensors and actuators to enable efficient line detecting and tracking.
- Completion of Design Reports 1 & .2
- Designing an appropriate chassis for the buggy.

Semester 2:

- Ensuring the buggy's robustness, reliability, and cost-effectiveness.
- Integrating a BLE Module into the buggy to enable the reception of remote commands for executing a 180° turn about its axis.
- Designing a PCB for improved functionality.
- Complete the four designated technical demonstrations.
- Completing the Proposal & Final reports [1]
- Additional: As a group, aiming to develop a high-speed buggy capable of participating in the race and achieving a competitive edge.

The adherence to these objectives provided a clear roadmap for the project's progression, aligning the team's efforts towards successfully realising an autonomous line-following buggy with enhanced features and optimal performance.

3.2. Changed Objectives during the Project and how they affected project planning and organisation of the team

During the project, several challenges were encountered, particularly related to the buggy's wheels, which significantly hindered the overall progress. Consequently, the buggy's performance on the heats was compromised, necessitating a reassessment of the initial objective of reaching race day. The team endeavoured to address and resolve issues to ensure the buggy could complete the track in a single run.

Despite the setback, most pre-established objectives remained unchanged throughout the project. The team accomplished the stated objectives during the heats, except for reaching race day. The project plan and team organisation remained consistent, as discussed earlier. To ensure equitable distribution of tasks,

the team considered individual timetables and impending coursework deadlines, allowing work to be allocated fairly and completed to a satisfactory level, particularly during high-pressure periods. Communication was crucial in maintaining coordination, primarily through a designated WhatsApp group chat, facilitating updates, feedback and problem resolution.

Furthermore, before each lab session, the team utilised the group chat to determine the attendance requirements for the day. However, considering the challenges faced with the buggy, adjustments were made to the team's organisation to sustain progress. Some team members not initially assigned to hardware-related tasks had to step in and resolve the encountered problems. While these changes were necessary to complete the buggy, they increased the workload for specific team members, indicating potential shortcomings in allocating and planning work. This aspect could have been addressed more efficiently through some forethought to mitigate the strain on team members.

In conclusion, the project experienced considerable difficulties, primarily associated with the buggy's wheels, leading to adjustments in the initial objectives and impacting the buggy's performance during the heats. However, the team demonstrated resilience and commitment in addressing the challenges. The allocation of tasks, through considering individual circumstances, could have been further refined to ensure a balanced workload. Overall, effective communication and adaptability were pivotal in overcoming obstacles and achieving most of the predetermined objectives, highlighting areas for improvement in future team collaborations.

3.3. Changes to the Gantt chart during the project

The hardware issues encountered during the project directly influenced the Gantt chart and required adjustments. Specifically, the setbacks related to the buggy's wheels resulted in a shift and extension of internal deadlines initially planned and documented in the Gantt Chart. This alteration allowed for sufficient time to address the hardware challenges and ensure the completion of the project tasks.

The changes to the Gantt chart reflected the need to allocate additional time for resolving unforeseen problems. By extending the deadlines, the team aimed to create a more realistic timeline that accounted for the unforeseen obstacles encountered. These adjustments were necessary to accommodate the extra work required for troubleshooting and rectifying the hardware problems, ultimately ensuring the successful progression of the project.

Moreover, the issues with the speed control stemming from the wheel difficulties led to re-evaluating the task dependencies and sequencing in the Gantt Chart. Specific tasks had to be revisited and potentially reordered to ensure the smooth flow of progress. This adjustment was essential to address the impact of the wheel issues on subsequent tasks and maintain a logical sequence of work.

Additionally, as the team became more aware of the workload demands from other modules, further modifications to the Gantt chart were made to balance the tasks and responsibilities effectively. This reallocation of work ensured that team members could manage their workload efficiently and simultaneously meet the requirements of multiple modules. By adjusting the task allocation within the Gantt chart, the team aimed to optimise productivity and maintain a harmonious integration of project work with academic commitments.

3.4. Adherence to plan and achievement of deliverables

The project did not unfold entirely as planned, primarily due to the emergence of issues related to the buggy's wheels. Consequently, the buggy's performance fell below the desired level for technical demonstrations 3 and 4. The presence of hardware issues hindered the achievement of optimal deliverables, necessitating the overcoming of numerous challenges along the way. Despite these obstacles, by the project conclusion, the buggy could fulfil all the predetermined deliverables, albeit not in the initially envisioned manner. Overcoming the hurdles required concerted efforts and problem-solving strategies.

Moreover, during the Easter break, the team members faced a substantial workload comprising lab reports for other modules. Unfortunately, this workload and logistical constraints made it infeasible for the group to convene during this period. Consequently, the progress initially planned to be accomplished over easter could not be fully realised. This created complications, as the technical demonstration was scheduled for immediately after Easter break, resulting in a significant surge in workload for preparing the buggy in the days leading to the demonstration. The time constraints and increased pressure during this critical phase adversely affected the planned preparations, with unforeseen issues arising on the buggy just before the demonstration.

3.5. Difficulties within the group and how they were resolved

The team encountered several difficulties throughout the project, with the primary challenge being the persistent issues related to the buggy wheels. These problems caused frustration within the group, but team members remained committed to resolving the issue and overcoming the obstacles despite the setbacks. Adjustments to the work allocation were made to distribute the demands of the buggy evenly among group members, ensuring that no individual became overwhelmed with the workload. This collaborative effort and task allocation flexibility helped alleviate the strain on team members.

Another significant challenge arose when preparing the buggy for technical demonstration 3, which was scheduled immediately after the Easter holidays. To address this, the group strategically utilised allocated lab sessions before and after Easter to complete as much work on the buggy as possible within the designated time. Additionally, Faisal took the initiative to bring the buggy home over Easter, allowing work to continue the project during the break. Regular progress updates were shared with the group to ensure all members remained informed and involved in the development.

Furthermore, the overlapping demands of other modules added to the complexity and stress experienced by the group. While this issue was anticipated from the beginning, as the semester progressed, the demands of other modules potentially increased beyond initial expectations. To mitigate the impact of these conflicting demands, weekly messages were posted on the group chat to inform members about their availability for upcoming lab sessions. This proactive communication and coordination enabled the distribution of work in a way that alleviated stress and reduced the burden on individual team members.

Despite the team working difficulties faced throughout the project, the group demonstrated resilience and a collaborative spirit in resolving issues. Effective communication, adjustments to work allocation, and optimising available resources were critical strategies employed to overcome challenges. The team's ability to adapt and find solutions amidst adversity showcases their dedication and commitment to completing the project.

3.6. Improvements to teamwork if the project were to be repeated

To further enhance teamwork, several additional measures could be implemented. Firstly, improving communication and documentation practices would be beneficial. Regular updates on group chats, GitHub repositories, and other shared platforms should be encouraged to ensure all team members are informed about the project's progress. This would facilitate better coordination and minimise last-minute explanations or confusion during technical demonstrations.

Secondly, increasing the frequency of team meetings in Semester 2 would allow for more inclusive decision-making. A sense of ownership and involvement can be fostered by involving all group members in discussions regarding allocating tasks and responsibilities for the technical demonstration. This collaborative approach ensures that everyone has a voice and contributes to all aspects of the project.

Furthermore, promoting cross-training and knowledge sharing among team members can enhance problem-solving abilities and overall understanding of the buggy. While individuals may have preferred focus areas, encouraging everyone to participate in the project's hardware and software aspects can provide a broader skill set and facilitate effective troubleshooting when issues arise.

By implementing these improvements, including better communication and documentation practices, more inclusive decision-making processes, and promoting cross-training, the overall teamwork within the group can be strengthened. These measures will improve coordination, knowledge sharing, and problem-solving capabilities, ultimately leading to more prosperous and cohesive project execution.

4. Budget vs. Outturn

The following section compares the budgeted and actual expenditures, providing an overview of the financial performance and variances within the project duration.

4.1. Discussion of Expenditures

Table 1 - Total amount spent during the project [1]

Component	Units	Seller	Cost Total (excl. VAT)
TCRT5000	4	RS	£ 0.92
4-Way SIL header	7	RS	£ 1.89
Sensor Mini PCB	3	UoM	£ 3.00
8 Pin IC DIP Socket	5	RS	£ 2.69
14 Pin IC DIP Socket	3	RS	£ 0.87
MCP6004	2	RS	£ 1.10
ICL-7660	1	RS	£ 1.90
MPU-6050	1	Farnell	£ 7.99
Total spent	£ 20.36		

Table 1 outlines the expenditures associated with various components, including the seller, unit quantities, and total costs.

Additional TCRT5000 units were purchased due to the limited availability of freely provided units. DIL headers were acquired to ensure the protection of chips on the board by avoiding direct soldering. The MCP6004 and ICL-7660 serving as an op-amp and analogue amplifier, respectively, were essential for implementing the error conversion logic of the PCB. The MPU-6050, a magnetometer, was included as a contingency measure in case the speed controller malfunctioned, enabling the inclination angle measurement to implement a predetermined speed. However, the speed controller worked perfectly, and thus the MPU-6050 was never utilised.

The total expenditure amounted to £20.36, utilising 50 % of the allocated budget of £40.

4.2. Retail price of the buggy

The autonomous line-following buggy is a valuable product with broad applications in the education industry, capable of enhancing students' interest in electronic and electrical engineering (EEE). As part of the project's evaluation, it is necessary to determine a suitable retail price that covers the production costs and allows for a reasonable profit margin.

By analysing the cost of each component used in the construction of the buggy, as detailed in [1], the total value of the buggy is determined to be £274.77. This value represents the costs of various components, including sensors, PCBs, Motors, and other necessary elements.

To establish a reasonable retail price, conducting a market survey becomes essential. This step involves researching the current market prices of similar products and analysing consumer preferences [4]. Understanding the market dynamics and competitive landscape will enable a more informed decision regarding the retail price.

The autonomous line-following buggy holds substantial commercial value due to its applications in the education industry. It can significantly contribute to students' learning experiences in EEE, expanding their knowledge and fostering their interest in the field and demonstrating the applications of such technology in the industry. The buggy's potential to meet educational objectives creates a demand that justifies a higher retail price.

Production costs play a significant role in determining the retail price. The extensive labour and testing involved in constructing the buggy should be considered. With over 50 hours dedicated to labs, designing and assembling parts, programming and testing, the associated production costs contribute to the product's overall value.

Considering the total value of the components. Market research, production costs, and the educational benefits offered by the autonomous line-following buggy, a reasonable retail price can be calculated. Applying a 15% profit margin [4] to the total value of the buggy (£274.77) results in a retail price of approximately £316.99. This price allows for a reasonable profit while considering the market value, production costs, and the educational advantages offered by the product. However, further market research and analysis may be necessary to ensure competitiveness and adjust the retail price accordingly.

By striking the right balance between affordability and the value proposition, the determined retail price will enable the successful dissemination of the autonomous line-following buggy to the target audience.

5. Analysis of Heats

This section comprehensively evaluates the autonomous line-following buggy's performance, focusing on its ability to accurately and efficiently follow a predefined path. It will highlight strengths, weaknesses, and insight for future improvements.

5.1. Preparation before the heats

In anticipation of the heat, comprehensive preparations were undertaken to ensure the buggy's optimal performance. The first crucial step involved verifying the functionality of the system's hardware. Meticulous checks were performed on the wire connections, and rigorous sensor testing/calibration was conducted to ensure their proper operation. Subsequently, extensive testing was carried out on the motor system, encompassing the wheels, gears, and motors.

A MATLAB simulation program was developed to determine the most suitable PID values to enhance the line-following capabilities. A toolbox was imported to facilitate motor system monitoring, and control system theories were applied to establish a second-order transfer function for the system. This approach significantly streamlined the controller's tuning process, ensuring optimal performance during line following. The image below illustrates the model used to obtain the PID parameters.

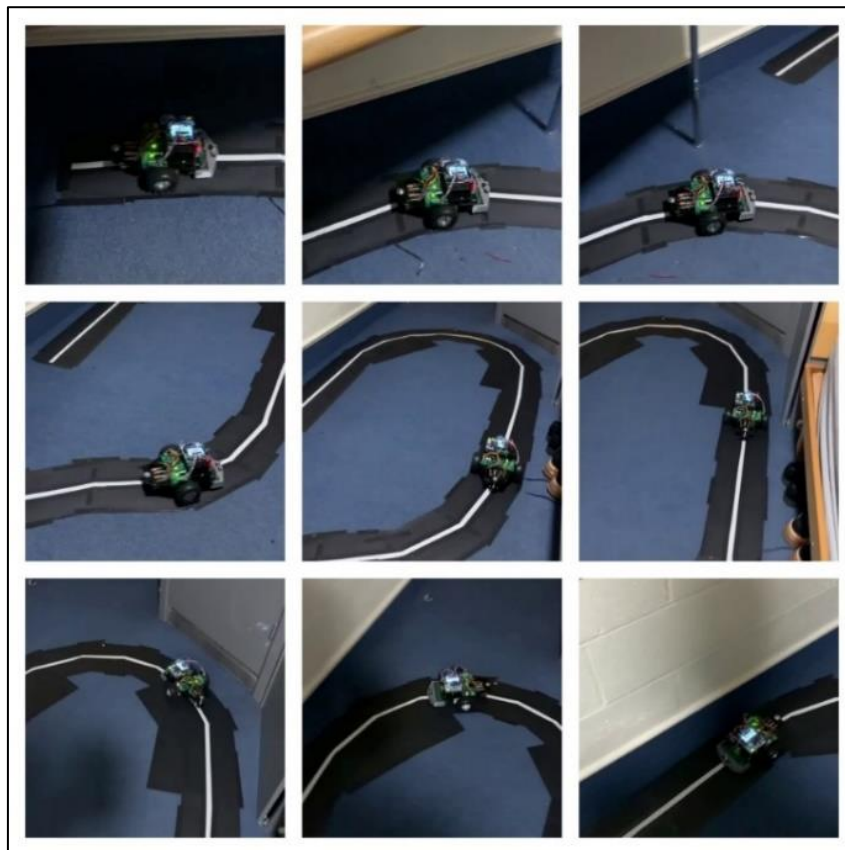


Figure 5 - Proof of testing on homemade track

Figure 5 shows the proof of testing the buggy on a homemade track.

In addition to software considerations, comprehensive testing was conducted to validate the buggy's readiness for the heat. The testing encompassed various manoeuvres such as navigating bends, traversing slopes, executing turnarounds and performing a controlled stop. These assessments were conducted on a track constructed at Faisal's accommodation, designed to replicate the required testing conditions. The outcomes of these rigorous tests provided a high level of confidence in the buggy's capabilities and confirmed its preparedness for the upcoming heats.

Overall, meticulous attention was given to hardware verification, software optimisation, and testing to ensure the buggy's reliability and performance. The extensive preparations underscored the team's commitment to delivering a competent and competitive entry for the heats.

5.2. Performance of Buggy in heats

The buggy's performance during the heats exhibited a series of challenges and subsequent improvements. In the initial attempt, the buggy encountered difficulties, initially stalling at the starting point and requiring three subsequent attempts to commence following the line. Further complications arose when the buggy ultimately failed to ascend a hill, reaching the top before stopping and sliding back, resulting in an unsuccessful outcome.

Analysis conducted during the intermission between heats identified a significant factor contributing to the problems – specifically, the threshold value for the buggy to halt at the endpoint was too low. Consequently, the buggy would abruptly stop after moving, incorrectly assuming the endpoint had been reached at specific points on the track. Adjustments were made to increase the threshold value during the break, enabling the completion of the line-following despite the potential consequence of not stopping precisely at the end-of-line. Additionally, the line sensor reading algorithm was found to be too sensitive for the track. This sensitivity rendered even slightly larger threshold values liable to halt the buggy during line following. The line sensors' high sampling frequency was suspected of contributing to the issue.

Despite an increased threshold value, the buggy encountered a new setback during the second attempt. It continually rotated at the starting point, necessitating four restarts without success. Subsequent analysis attributed this to an incorrect wire connection. Due to time constraints and a hurried reconnection of wires to their correct positions, an error occurred. Specifically, the bipolar wire for the left wheel was erroneously connected to the wrong pin. Consequently, the motor exhibited erratic movements in direction and speed, resulting in the immediate failure of the second attempt.

Undeterred, the team persevered and completed all heats sections in the third and final attempt. Unfortunately, the wiring issue was not corrected until after the heats; thus, the buggy started spinning just before it could reach the end of the line point. This resulted in a 10-mark penalty.

The subsequent adjustments and troubleshooting efforts demonstrated the team's determination and resilience in overcoming challenges and improving the buggy's performance throughout the heats.

5.3. Changes made between heats and performance of changes

In response to the challenges faced during the heats, a series of modifications were implemented to rectify the issue and enhance the functionality of the buggy. One significant change involved adjusting the sensitivity of the sensors by raising the threshold value used to detect the end of the line. The adjustment effectively mitigated the problem of sudden stops caused by erroneous detection of the line's termination point.

Additionally, an attempt was made to address the issue of continuous spinning by reducing the buggy's speed. However, this alteration did not yield the desired outcome, as the root cause of the problem was the incorrect wiring of the left motor. The incorrect connection of wires resulted in erratic behaviour, rendering the speed adjustment ineffective in resolving the underlying issue.

These modifications aimed to improve the buggy's performance and address the specific challenges encountered during the heats. Despite the unsuccessful attempt to reduce the speed, the adjustments made to the sensitivity proved crucial in enabling the successful completion of the line-following objective.

5.4. Successful/ Worst Features

The buggy exhibits several successful features that contribute to its overall performance. Firstly, implementing hardware processing for analogue values from the line sensors on the PCB enhances efficiency by reducing sampling and processing times. This approach optimises the utilisation of STM32's ADC, resulting in improved data handling.

Secondly, the buggy incorporates a well-tuned PID controller for speed regulation, ensuring rapid attainment of the desired speed while minimising oscillations. Utilising a MATLAB simulation program aided in precisely determining PID gains for the line following controller, significantly reducing the time required for manual tuning. Additionally, introducing an LPF minimised sampling noise and facilitated the smooth movement of the buggy.

Furthermore, the algorithm employed by the group exhibits exceptional energy efficiency, enabling prolonged battery life, with charging required only every two days. This efficiency contributes to the overall effectiveness of the buggy's operation.

Lastly, incorporating a sensor cover shield, produced using 3D printing technology, protects against ambient light interference, direct light exposure, and potential sensor damage. This shield enhances the reliability and robustness of the sensor system.

However, the buggy experiences a notable weakness concerning the left wheel, manifesting as persistent issues throughout its operation. This problem initially emerged during TD3 and hindered the buggy's ability to ascend the uphill section. Despite resolving the issue one week before the heats, the left wheel ceased functioning again after a weekend. Consequently, the left wheel remained completely stationary shortly before the heat. The misconnection of the motor drive board, occurring during the fault-finding process, emerged as the primary cause of the group's failure during the heats.

In addition, due to an incorrect design in the chassis, the weight distribution was not as initially envisioned. To compensate for this, a castor wheel was added in front of the buggy. This increased the length of the buggy considerably and limited the buggy's ability to turn on sharp corners.

Overall, while the buggy demonstrates commendable features such as hardware processing, PID controller optimisation, energy efficiency, and sensor protection, the recurring problems with the left wheel and the misconnection of the motor drive board have posed significant challenges and affected its performance in critical moments.

6. References

- [1] ESP Group 25, "Proposal Report," The University of Manchester, Manchester, 2023.
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